

Math Camp

Lecture 8

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Approximating a function

Stone-Weierstrass theorem

- Suppose $K \subset \mathbb{R}^n$ is compact and $f : K \rightarrow \mathbb{R}$ is continuous
- f can be approximated arbitrarily well by a polynomial
 - Take the normed vector space $(C(K), \|\cdot\|_\infty)$, where $f \in C(K)$
 - For any $\varepsilon > 0$ there is a polynomial $p \in C(K)$ such that
$$\|f - p\|_\infty < \varepsilon$$
- Polynomials can be evaluated by computers, but the theorem doesn't determine the approximation
 - It is an existence theorem

Taylor's theorem

- If we additionally restrict to differentiable functions, and only require *local* approximations, Taylor's theorem provides the construction we are looking for
- Let $U \subset \mathbb{R}$ be open and $f : U \rightarrow \mathbb{R}$ be C^2 . For a given $x^* \in U$,

$$f(x) = f(x^*) + f'(x^*)(x - x^*) + \mathcal{O}(|x - x^*|^2)$$

- $\mathcal{O}$, pronounced “Big Oh”, is a special terminology for error functions
 - Think of $\mathcal{O}(y)$ as a function that is the same “size” as y
 - If y is small then $\mathcal{O}(y)$ is small

Taylor's theorem

- If x is near x^* then $|x - x^*|^2$, and thus $O(|x - x^*|^2)$, is very nearly zero, so

$$f(x) \approx f(x^*) + f'(x^*)(x - x^*)$$

- This is the *first-order* (or linear) approximation of f around x^* and is a polynomial of degree one
- Taylor's theorem makes formal the sense in which a differentiable function is locally linear

Taylor's theorem

- Suppose f is C^3 . Then

$$f(x) = f(x^*) + f'(x^*)(x - x^*) + \frac{1}{2}f''(x^*)(x - x^*)^2 + \mathcal{O}(|x - x^*|^3)$$

- ▶ Without the error term the RHS is the *second-order* approximation, and is a polynomial of degree 2
- ▶ If x is sufficiently near x^* it will necessarily be better than the first-order one
- ▶ Intuitively, the second-order term partially corrects for the fact that f' varies with x

Taylor's theorem

- The general pattern can now be recognized
 - If f is C^{N+1} then

$$f(x) = f(x^*) + \sum_{n=1}^N \frac{f^{(n)}(x^*)}{n!} (x - x^*)^n + \mathcal{O}(|x - x^*|^{N+1})$$

- Provided a function is sufficiently smooth, Taylor's theorem provides a polynomial approximation to arbitrary accuracy

Taylor's theorem

Taylor's expanse of e^x around the origin

Taylor's theorem: Multivariate

- Let $U \subset \mathbb{R}^n$ and assume f is C^2
- The first-order approximation of f near $\mathbf{x}^*$ is the direct analog

$$f(\mathbf{x}) = f(\mathbf{x}^*) + Df(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) + \mathcal{O}(\|\mathbf{x} - \mathbf{x}^*\|^2)$$

- ▶ $Df(\mathbf{x}^*)$ may be viewed as a row vector, and as such it is the linear functional taking changes in $\mathbf{x}$ to changes in f

Taylor's theorem: Multivariate

- Assume now f is C^3 . To obtain the second-order approximation we must differentiate $Df : U \rightarrow \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$
- The *Hessian* $D^2 f$ of f at $\mathbf{x}^*$ is the $n \times n$ symmetric matrix of second partials, $D^2 f(\mathbf{x}^*) = (f_{ij}(\mathbf{x}^*))$

$$D^2 f = \begin{pmatrix} f_{11} & \cdots & f_{n1} \\ \vdots & \cdots & \vdots \\ f_{1n} & \cdots & f_{nn} \end{pmatrix}$$

- Taylor's theorem says

$$\begin{aligned} f(\mathbf{x}) &= f(\mathbf{x}^*) + Df(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) \\ &\quad + \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)' D^2 f(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) + \mathcal{O}(\|\mathbf{x} - \mathbf{x}^*\|^3) \end{aligned}$$

Taylor's theorem: Multivariate

- The second-order term is captured by the Hessian acting as a quadratic form
 - As in the univariate case, it corrects for the fact that Df is not constant
- Written out,

$$(\mathbf{x} - \mathbf{x}^*)' D^2 f(\mathbf{x}^*) (\mathbf{x} - \mathbf{x}^*) = \sum_{ij} f_{ij}(\mathbf{x}^*) (x_i - x_i^*) (x_j - x_j^*)$$

- If $\mathbf{x}$ moves a little along the i th coordinate then f moves by $df = f_i(\mathbf{x}^*) dx_i$
- But if $\mathbf{x}$ also moves in the j th direction then $f_i(\mathbf{x}^*) dx_i$ changes by $f_{ij}(\mathbf{x}^*) dx_i dx_j$, so we should adjust for this

The implicit function theorem

- The implicit function theorem is useful to assess comparative statics experiments
- Consider an *economic model* with a triple $(\mathbf{x}, \mathbf{y}, f)$ with
 - $\mathbf{x} \in \mathbb{R}^m$ the *exogenous* variables, taken as given by the model
 - $\mathbf{y} \in \mathbb{R}^n$ the *endogenous* variables, pinned down by the model
 - $f : \mathbb{R}^m \oplus \mathbb{R}^n \rightarrow \mathbb{R}^n$ capturing the equilibrium restrictions

The implicit function theorem

- We take the values of $\mathbf{x}$ as given and use the n equations $f(\mathbf{x}, \mathbf{y}) = 0$ to determine the values of $\mathbf{y}$
 - f defines the y 's as implicit functions of the x 's
 - The equilibrium restrictions are the mechanism through which we may write the endogenous variables as implicit functions of the exogenous ones

The implicit function theorem

- Let $D(p, I)$ be quantity demanded given price p and income I , and $S(p)$ quantity supplied
- Define the *excess demand* function $Z(p, I) = D(p, I) - S(p)$
 - I is exogenous, p is endogenous, and $Z(p, I) = 0$ is the equilibrium restriction
- Since f is nonlinear, we can prove existence of a unique equilibrium, but we might not be able to explicitly solve for it
- By computing total differentials we can solve for the *changes* in the endogenous variables in terms of the *changes* in the exogenous variables
 - The total differential is linear in the differential changes

Comparative statics

- What happens to p if I increases?
 - Writing p^* as the equilibrium value, $p^* = p^*(I)$, we can compute $\frac{\partial p^*}{\partial I}$

- Take the total differential of Z ,

$$dZ = Z_p dp + Z_I dI$$

- The system must remain in equilibrium, so $dZ = 0$ and

$$Z_p dp + Z_I dI = 0 \quad \text{or} \quad \frac{\partial p^*}{\partial I} = -\frac{Z_I}{Z_p}$$

Comparative statics

- Noting $Z_I = D_I > 0$ and $Z_p = D_p - S_p < 0$,

$$\frac{\partial p^*}{\partial I} = \frac{D_I}{S_p - D_p} > 0$$

- ▶ An increase in income leads to an increase in equilibrium price

Comparative statics

$$a - by = c\sqrt{y - x}$$

Comparative statics

- The general case follows the same pattern: if $\mathbf{x}$ changes by $d\mathbf{x}$, how must y change so that the model remains in equilibrium?
- Compute total differentials again and set the changes to zero

$$df = D_{\mathbf{x}}f d\mathbf{x} + D_yf dy = 0$$

and solve for dy in terms of $d\mathbf{x}$

- ▶ $D_{\mathbf{x}}f$ is the $n \times m$ matrix of partials of the f^i with respect to $x_1, \dots, x_m$

$$D_{\mathbf{x}}f(x, y) = \begin{pmatrix} f_{x_1}^1(x, y) & \dots & f_{x_m}^1(x, y) \\ \vdots & \dots & \vdots \\ f_{x_1}^n(x, y) & \dots & f_{x_m}^n(x, y) \end{pmatrix}$$

Comparative statics

- $D_{\mathbf{y}}f$ is the $n \times n$ matrix of partials of the f^i with respect to $y_1, \dots, y_n$

$$D_y f(x, y) = \begin{pmatrix} f_{y_1}^1(x, y) & \dots & f_{y_n}^1(x, y) \\ \vdots & \dots & \vdots \\ f_{y_1}^n(x, y) & \dots & f_{y_n}^n(x, y) \end{pmatrix}$$

- We can “solve” for dy provided $\det D_{\mathbf{y}}f(\mathbf{x}, \mathbf{y}) \neq 0$

The implicit function theorem

Implicit Function Theorem

Suppose $f : \mathbb{R}^m \oplus \mathbb{R}^n \rightarrow \mathbb{R}^n$ has continuous first partials. If $f(\mathbf{x}^*, \mathbf{y}^*) = 0$ and $\det D_{\mathbf{y}} f(\mathbf{x}^*, \mathbf{y}^*) \neq 0$ then there exists an open set $U \subset \mathbb{R}^m$ with $\mathbf{x}^* \in U$, and a continuously differentiable function $g : U \rightarrow \mathbb{R}^n$ so that

- $\mathbf{y}^* = g(\mathbf{x}^*)$
- $\mathbf{x} \in U \implies f(\mathbf{x}, g(\mathbf{x})) = 0$
- $Dg(\mathbf{x}^*) = -D_{\mathbf{y}} f(\mathbf{x}^*, \mathbf{y}^*)^{-1} \circ D_{\mathbf{x}} f(\mathbf{x}^*, \mathbf{y}^*)$

The implicit function theorem

$\begin{pmatrix} 10-2p_1+p_2 \\ 8-2p_2+p_1 \end{pmatrix} = \begin{pmatrix} p_1-t \\ p_2 \end{pmatrix}$ where $p = (p_1, p_2)$ is endogenous and t is exogenous.

Inner products

- A norm provides a vector space with a notion of *distance*; an inner product imparts *geometry*
- If $\mathbf{V}$ is a real vector space then an *inner product* is a function $\langle \cdot, \cdot \rangle : \mathbf{V} \times \mathbf{V} \rightarrow \mathbb{R}$ which is
 - *Symmetric*, $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$
 - *Bilinear*, linear in both variables
 - *Positive definite*, $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ and $\langle \mathbf{v}, \mathbf{v} \rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$
- The associated geometry is provided by *orthogonality*: $\mathbf{v}$ and $\mathbf{w}$ are orthogonal if $\langle \mathbf{v}, \mathbf{w} \rangle = 0$

Inner products

$$\mathbf{v} = (3, 1) \text{ and } \mathbf{w} = (-1, 3)$$

$$\mathbf{v} = (3, 1) \text{ and } \mathbf{w} = (2, 3)$$

Inner products

- If $\mathbf{V} = \mathbb{R}^n$ the canonical inner product is the usual dot product,
 $\langle \mathbf{v}, \mathbf{w} \rangle = \mathbf{v}' \mathbf{w}$
 - If Q is any $n \times n$ positive definite matrix then $\mathbf{v}' Q \mathbf{w}$ also identifies an inner product
 - These quadratic forms characterize the collection of inner products on $\mathbb{R}^n$
- The norm induced by an inner product is $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle$
 - The norm induced by the canonical inner product on $\mathbb{R}^n$ is the usual Euclidean norm

Projection

- How to think about the general meaning of $\langle \mathbf{v}, \mathbf{w} \rangle$?
- Imagine $\mathbf{w}$ has unit length and let $\xi = \mathbf{v} - \alpha \mathbf{w}$ so that $\langle \mathbf{w}, \xi \rangle = 0$. Then $\mathbf{v} = \alpha \mathbf{w} + \xi$ and

$$\langle \mathbf{v}, \mathbf{w} \rangle = \alpha \langle \mathbf{w}, \mathbf{w} \rangle + \langle \xi, \mathbf{w} \rangle = \alpha$$

- Thus $\langle \mathbf{v}, \mathbf{w} \rangle$ identifies how far $\mathbf{v}$ extends in the direction determined by $\mathbf{w}$
 - It is the *projection* of $\mathbf{v}$ onto $\mathbf{w}$

Projection

- Suppose $B = \{\mathbf{b}_1, \dots, \mathbf{b}_m\} \subset \mathbf{V}$ and let $\mathbf{v} \in \mathbf{V}$
- Our goal is to find $\mathbf{w} \in \text{span}(B)$ so that $\mathbf{v} = \mathbf{w} + \xi$ with ξ orthogonal to each $\mathbf{b}_i$
 - We seek $\mathbf{w}$ so that $\langle \mathbf{b}_i, \mathbf{v} - \mathbf{w} \rangle = 0$ for each i
 - Writing $\mathbf{w} = \sum \alpha_i \mathbf{b}_i$,

$$\langle \mathbf{b}_i, \mathbf{v} - \mathbf{w} \rangle = 0 \implies \langle \mathbf{b}_i, \mathbf{v} \rangle = \sum_j \alpha_j \langle \mathbf{b}_i, \mathbf{b}_j \rangle$$

- Letting M be the $n \times m$ matrix having columns $\mathbf{b}_i$,

$$\alpha = (M' M)^{-1} M' \mathbf{v}$$

Projection

- Given a set of linearly independent vectors B we may always construct $\hat{B}$ so that B and $\hat{B}$ span the same space and the elements of $\hat{B}$ are orthogonal
- Then the columns of M are orthogonal, so $M' M$ is diagonal
 - A diagonal matrix is very easy to invert and has minimal conditioning number
 - The benefits are analogous to those arising from conditioning on independent regressors in a linear regression model